WHY PYBIO-10

# From the bench to the code.

Spreadsheets break when the data gets big. In 10 evenings you learn to clean, analyse, visualise and model your own biological data in Python, the language used across modern genomics, bioinformatics and lab analytics.

```python
# Day 7: your first sequence scan
from Bio import SeqIO
from Bio.SeqUtils import gc_fraction
for r in SeqIO.parse("genes.fasta", "fasta"):
    print(r.id, gc_fraction(r.seq))
```

## By Day 10 you will be able to

- Write clean, reusable Python in Jupyter and Google Colab
- Turn messy lab and field data into tidy, analysis-ready tables
- Run t-tests, ANOVA and regression, and make publication-ready figures
- Read FASTA and GenBank files, run BLAST, and handle gene-expression data
- Build and evaluate your first machine-learning model for biology

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

AVYAGRAHA RESEARCH AND ANALYTICS LLP PRESENTS

REGISTRATIONS OPEN

# PyBio-10

## Python for Life Sciences

10-DAY HANDS-ON TRAINING PROGRAMME

**DATES**
16 – 26 Nov 2026
Sunday 22 Nov holiday

**TIMING**
6:30 – 8:30 PM IST
2 hours per evening

**MODE**
Online · Live
Hands-on, no prior coding

**FEE (INCL. GST)**
₹1,499 / ₹1,999
Students / Professionals

**RECORDINGS**
3-month access
Revisit every session

**CERTIFICATE**
E-certificate
Plus capstone projects

Python · NumPy · Pandas · Matplotlib · Seaborn · SciPy · Biopython · scikit-learn

WHO CAN REGISTER

# Built for life scientists, not programmers.

- UG, PG and PhD students in the life sciences
- Faculty, scientists and research associates
- Biotechnology, Microbiology, Biochemistry, Genetics, Agriculture, Pharmacy and Medical Sciences
- Pharma, agri-biotech and healthcare professionals
- Anyone moving from Excel or SPSS to code-based analysis

**No prior programming experience needed.** A laptop with internet access and a Google account (for Colab) is all you need.

| 10 | 20 h | 11 |
|---|---|---|
| live evening sessions | of hands-on coding | tools and libraries |

**Start where you are.**
**Code your own data.**

ENROLL NOW

aviqlabs.com/pybio10

**Students**
UG / PG / PhD students and research scholars
**₹1,499**

**Professionals**
Faculty, scientists, industry and others
**₹1,999**

All fees inclusive of GST · Limited seats

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**Questions?**

Call or WhatsApp
**+91 83105 79116 · +91 94803 63456**
Email **avinashstat@aviqlabs.com**

COURSE STRUCTURE

# 10 days. 10 skills.

Every session is live and hands-on: you write and run the code yourself, on biological datasets.

## Module 1 · Python Foundry Days 1–2

**D01 Inauguration · Python foundations**
Colab notebooks, variables, data types, operators
Mon 16 Nov

**D02 Thinking in code**
Data structures, conditions, loops, functions
Tue 17 Nov

## Module 2 · Scientific Data Toolkit Days 3–5

**D03 NumPy for scientific computing**
Arrays, broadcasting, vectorised calculations
Wed 18 Nov

**D04 Pandas for lab data**
Clean, validate, merge and reshape datasets
Thu 19 Nov

**D05 Publication-ready visualisation**
Matplotlib, Seaborn, multi-panel figures
Fri 20 Nov

## Module 3 · Statistics and Sequences Days 6–7

**D06 Biostatistics in Python**
t-tests, ANOVA, regression, assumptions
Sat 21 Nov

**D07 Biopython: sequence analysis**
FASTA, GenBank, translation, BLAST
Mon 23 Nov

## Module 4 · Omics and ML Days 8–9

**D08 Omics data handling**
Expression matrices, FDR, volcano plots
Tue 24 Nov

**D09 Machine learning for biology**
Classification, evaluation, clustering, PCA
Wed 25 Nov

## Module 5 · Capstone Day 10

**D10 Capstone and valedictory**
End-to-end analysis, presentations, e-certificates
Thu 26 Nov

Sunday 22 November — HOLIDAY · NO SESSION — Every session: 6:30 – 8:30 PM IST

TOOLS YOU WILL USE: `Python 3` `Jupyter` `Google Colab` `NumPy` `Pandas` `Matplotlib` `Seaborn` `SciPy` `statsmodels` `Biopython` `scikit-learn`

COORDINATOR AND RESOURCE PERSONS

# Learn from practitioners.



COORDINATOR · DAYS 1, 6–10

### Dr. Avinash Gopa

Founder & CEO, Avyagraha Research & Analytics LLP · Ex-ICMR Project Scientist-II · Ph.D. (Agricultural Statistics), IARI, New Delhi



DAYS 1, 4 AND 5

### Mr. Chandan G Y

Data Analyst, Bangalore



DAYS 2 AND 3

### Mr. Rajesh A

CEO & Tech Member, Avinya Techsphere, Bangalore



DAY 6

### Mr. Shrey Pandya

Research Analyst, Population Research Centre, Department of Statistics, The Maharaja Sayajirao University of Baroda, Vadodara



DAY 9

### Dr. Sandip Garai

Scientist, ICAR-Indian Institute of Agricultural Biotechnology, Garhkhatanga, Ranchi

KEY TAKEAWAYS

# You walk away with

**.py — Code notebooks**
Every day's notebook, ready to reuse on your own research data

**3M — 3-month recordings**
Revisit any session at your own pace

**CP — Capstone projects**
End-to-end analyses you can show on your CV

**EC — E-certificate**
Issued on completion of the programme

**DNA — Real biological datasets**
Sequence, expression, lab and field data from published studies

**Q&A — Live doubt clearing**
Direct interaction with the trainers in every session

ABOUT AVYAGRAHA

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

A DPIIT-recognised deep-tech startup, incubated at AIC-UAS-B, GKVK, Bengaluru. We provide data-driven solutions, training programmes and research collaborations.

- **Training:** statistics, R, Python, AI/ML and Generative AI for institutions and professionals
- **Research consultancy:** study design, data analysis and statistical modelling
- **Partnerships:** MoUs with universities and R&D institutes

## Register now

Scan the QR, pay by UPI and upload your payment screenshot.

`aviqlabs.com/pybio10`

## Contact us


+91 83105 79116 · +91 94803 63456

avinashstat@aviqlabs.com

www.aviqlabs.com

DETAILED TRAINING PROGRAMME CURRICULUM

# PyBio-10: Python for Life Sciences
## 10-Day Hands-On Training Programme

**16 Nov 2026 – 26 Nov 2026 · 6:30 PM to 8:30 PM IST · Online (live)** · Sunday holiday

Organized by Avyagraha Research and Analytics LLP, Ramasagara, Ballari, Karnataka

www.aviqlabs.com · avinashstat@aviqlabs.com · +91 83105 79116 · +91 94803 63456

10 live sessions | 20 hours hands-on | 3-month recordings | Capstone projects | E-certificate

| Date | Topics | Resource person |
|---|---|---|
| **Module 1 · Python Foundry (Days 1–2)** | | |
| 16/11/2026<br>Monday | D01<br>**Inauguration and Python foundations**<br>Programme orientation and setup; notebooks, cells and Markdown; variables, naming, numbers, strings and Boolean values; operators, type conversion, print(), input() and simple errors. Practical: a documented notebook of sample IDs, treatment labels and measurements with a simple summary. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha<br>&<br>**Mr. Chandan G Y**<br>Data Analyst, Bangalore |
| 17/11/2026<br>Tuesday | D02<br>**Thinking in code: data structures and logic**<br>Lists, tuples, dictionaries and sets; if / elif / else conditions; for and while loops over samples; functions, parameters, return values and basic debugging. Practical: count DNA bases or codons, flag out-of-range measurements and convert assay units with a reusable function. | **Mr. Rajesh A**<br>CEO & Tech Member, Avinya Techsphere, Bangalore |
| **Module 2 · Scientific Data Toolkit (Days 3–5)** | | |
| 18/11/2026<br>Wednesday | D03<br>**NumPy for scientific computing**<br>Arrays, shape, dimensions, data types and indexing; slicing, reshaping and broadcasting; vectorised arithmetic and axis-wise aggregation; random numbers, a simple simulation and seeds for reproducibility. Practical: process a plate-reader absorbance matrix with means, SDs and blank correction. | **Mr. Rajesh A**<br>CEO & Tech Member, Avinya Techsphere, Bangalore |
| 19/11/2026<br>Thursday | D04<br>**Pandas for biological and laboratory data**<br>Series and DataFrames; reading CSV/Excel; data types, duplicate records, missing values and basic validation; filtering, sorting, grouping and summary tables; merge, groupby, pivot and melt for tidy data. Practical: clean a lab dataset, join it with sample metadata and keep a data-quality log. | **Mr. Chandan G Y**<br>Data Analyst, Bangalore |
| 20/11/2026<br>Friday | D05<br>**Publication-ready scientific visualisation**<br>Figure, axes, labels, legends, scales and annotations; box, violin and scatter plots, regression lines and heatmaps; error bars and communicating variation; multi-panel layouts and export to high-resolution PNG, TIFF or PDF. Practical: two labelled figures for a treatment-versus-control dataset. | **Mr. Chandan G Y**<br>Data Analyst, Bangalore |
| **Module 3 · Statistics and Sequences (Days 6–7)** | | |
| 21/11/2026<br>Saturday | D06<br>**Biostatistics in Python**<br>Descriptive statistics and exploratory checks; normality, variance and study-design assumptions; independent/paired t-tests and chi-square tests; one-way/two-way ANOVA with Tukey post-hoc; correlation and linear regression with SciPy and statsmodels. Practical: a short statistical report on a treatment-control dataset. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha<br>&<br>**Mr. Shrey Pandya**<br>Research Analyst, Dept. of Statistics, MSU Baroda, Vadodara |
| 22/11/2026<br>Sunday | HOLIDAY | |

| Date | Topics | Resource person |
|---|---|---|
| 23/11/2026<br>Monday | D07<br>**Biopython essentials: sequence analysis**<br>Seq objects and DNA, RNA and protein basics; GC fraction, transcription and translation; parsing FASTA and GenBank with SeqIO; NCBI Entrez and responsible use of public records; pairwise alignment and BLAST output. Practical: a reusable sequence-analysis notebook. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha |
| **Module 4 · Omics and Machine Learning (Days 8–9)** | | |
| 24/11/2026<br>Tuesday | D08<br>**Omics data handling**<br>Gene-expression matrices and sample metadata; FASTQ, SAM/BAM and VCF formats; counts versus normalised expression and log transformation; fold change, p-values and false discovery rate (FDR); volcano plots and clustered heatmaps. Practical: explore a prepared expression dataset and visualise results. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha |
| 25/11/2026<br>Wednesday | D09<br>**Machine learning for biology**<br>Features, target, train/test split and preprocessing; logistic regression and random forest; confusion matrix, precision, recall, ROC curve and class imbalance; k-means and hierarchical clustering; PCA, data leakage and overfitting. Practical: train and evaluate a baseline classifier on held-out data. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha<br>&<br>**Dr. Sandip Garai**<br>Scientist, ICAR-IIAB, Ranchi |
| **Module 5 · Capstone (Day 10)** | | |
| 26/11/2026<br>Thursday | D10<br>**Capstone, certification and valedictory**<br>Capstone workflow: question, data, quality checks, analysis and visualisation; reproducibility with clear cells, comments and fixed seeds; short presentations, feedback and valedictory. Output: a runnable notebook, a labelled figure and a brief summary with limitations. | **Dr. Avinash Gopa**<br>Founder & CEO, Avyagraha |

## Programme fees

| Category | Fee (incl. GST) |
|---|---|
| **Students**: UG / PG / PhD and research scholars | **Rs. 1,499 /-** |
| **Professionals**: faculty, scientists, industry, others | **Rs. 1,999 /-** |

## Requirements

- A laptop or desktop with a stable internet connection.
- A Google account, for Google Colab notebooks.
- No software to install: everything runs in the browser.

## How to register

1. Scan the QR code or visit **aviqlabs.com/pybio10**.
2. Tap **Pay now** (any UPI app opens with the fee filled in), or pay to UPI ID **9480363456@okbizaxis** (Avyagraha Research and Analytics LLP).
3. Note the **UPI transaction ID** shown after payment.
4. Enter the transaction ID, **upload the payment screenshot** and submit to get your registration ID by email.
5. Joining details will be shared by email and WhatsApp before the programme begins.

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